

01 Introduction

The Problem

- Passwords and SMS-OTP fail to phishing, credential stuffing and SIM-swap.
- Face login is spoofable by printed photos, screen replay, masks and deepfakes.
- Every app rebuilds auth from scratch, and KVKK / GDPR compliance gets tacked on afterwards.
- Mainstream IAM (Okta, Auth0, Azure AD) treat biometrics as a device-local feature.

The project aims to develop a **multi-platform, multi-tenant, and cloud-native SaaS platform** for secure and flexible **multi-biometric identity authentication**, with KVKK / GDPR compliance built in.

What FIVUCSAS Is

A multi-tenant biometric identity SaaS that is already running in production. Every login is confirmed with **your own biometric data**, behind a single **"Sign in with FIVUCSAS"** button. It plugs into any app the way e-Devlet or "Sign in with Google" do, only here the identity proof is biometric, and you can host the whole thing yourself.

- **Face recognition:** Facenet-512 embeddings, pgvector IVFFlat, 1:1 and 1:N.
- **Hybrid liveness:** the active Biometric Puzzle plus passive MiniFASNet.
- **Ten factors:** password, OTPs, TOTP, QR, face, voice, WebAuthn, FIDO2, NFC.
- **Multi-platform:** Android (KMP) and a React 18 admin.

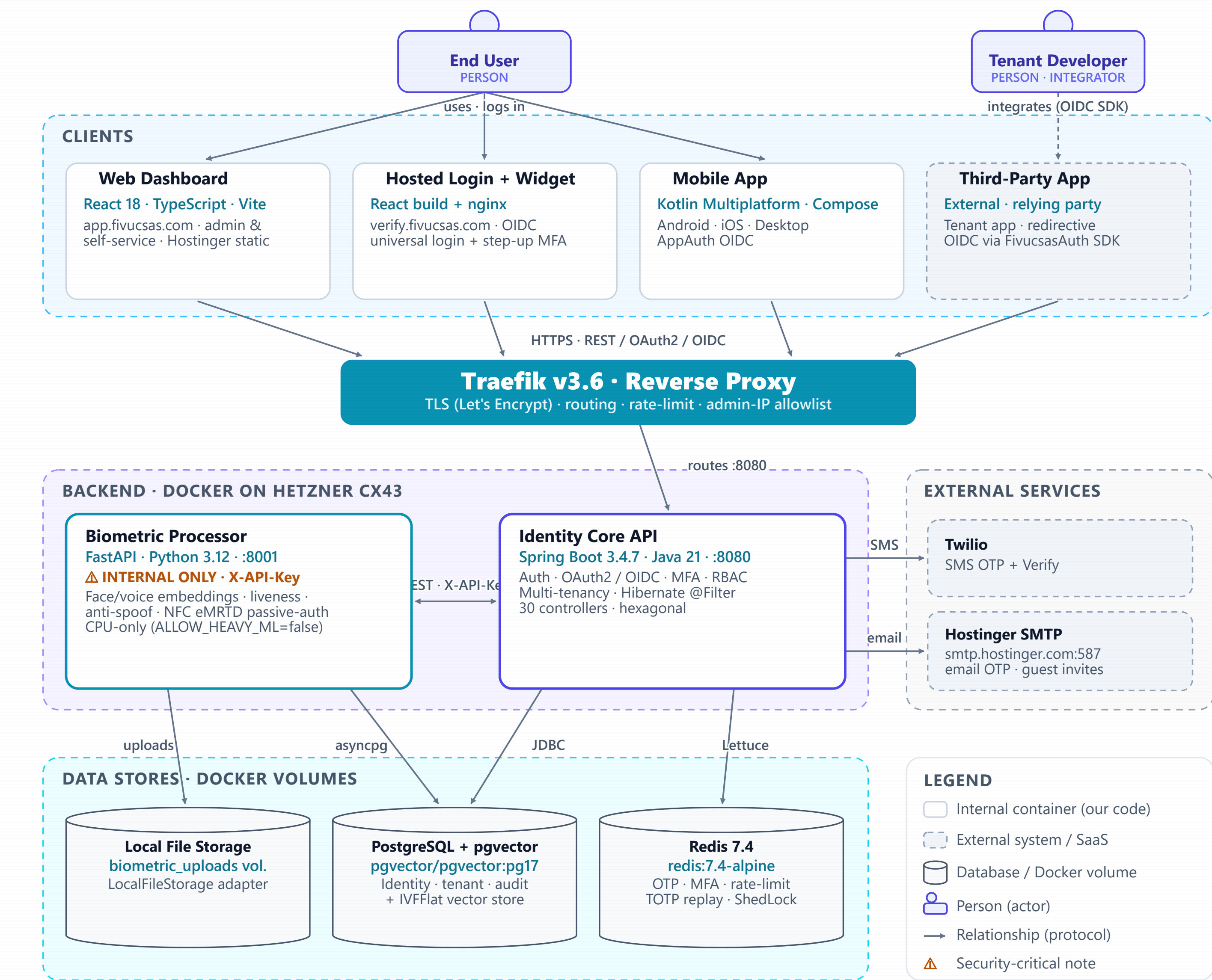
OBJECTIVES

Face Recognition Sub-second 1:1 / 1:N on commodity CPU.	Biometric Puzzle Randomised active liveness against photo, replay and deepfake.	Multi-Tenant SaaS Ten factors; per-tenant MFA flows by JSON, not Java.	Hybrid Anti-Spoofing Active puzzle plus passive PAD; ISO/IEC 30107-3 aligned.
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02 Methodology and Technical Approach

Container Diagram

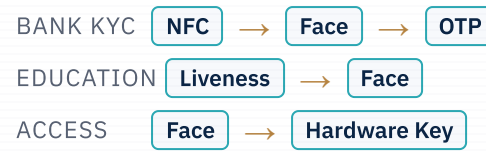
Multi-tenant biometric authentication platform. Clients reach a single Traefik edge; only the Identity Core API is public; the Biometric Processor is internal-only (Docker network + X- API-Key), and the two external comms providers are called solely by the Spring API.



TENANT-CONFIGURABLE VERIFICATION

One platform, infinite flows

Each tenant builds its own flow: which steps run, in what order, and with which thresholds. Everything is assembled at runtime, with no code changes.



GPU-FREE FACE PIPELINE

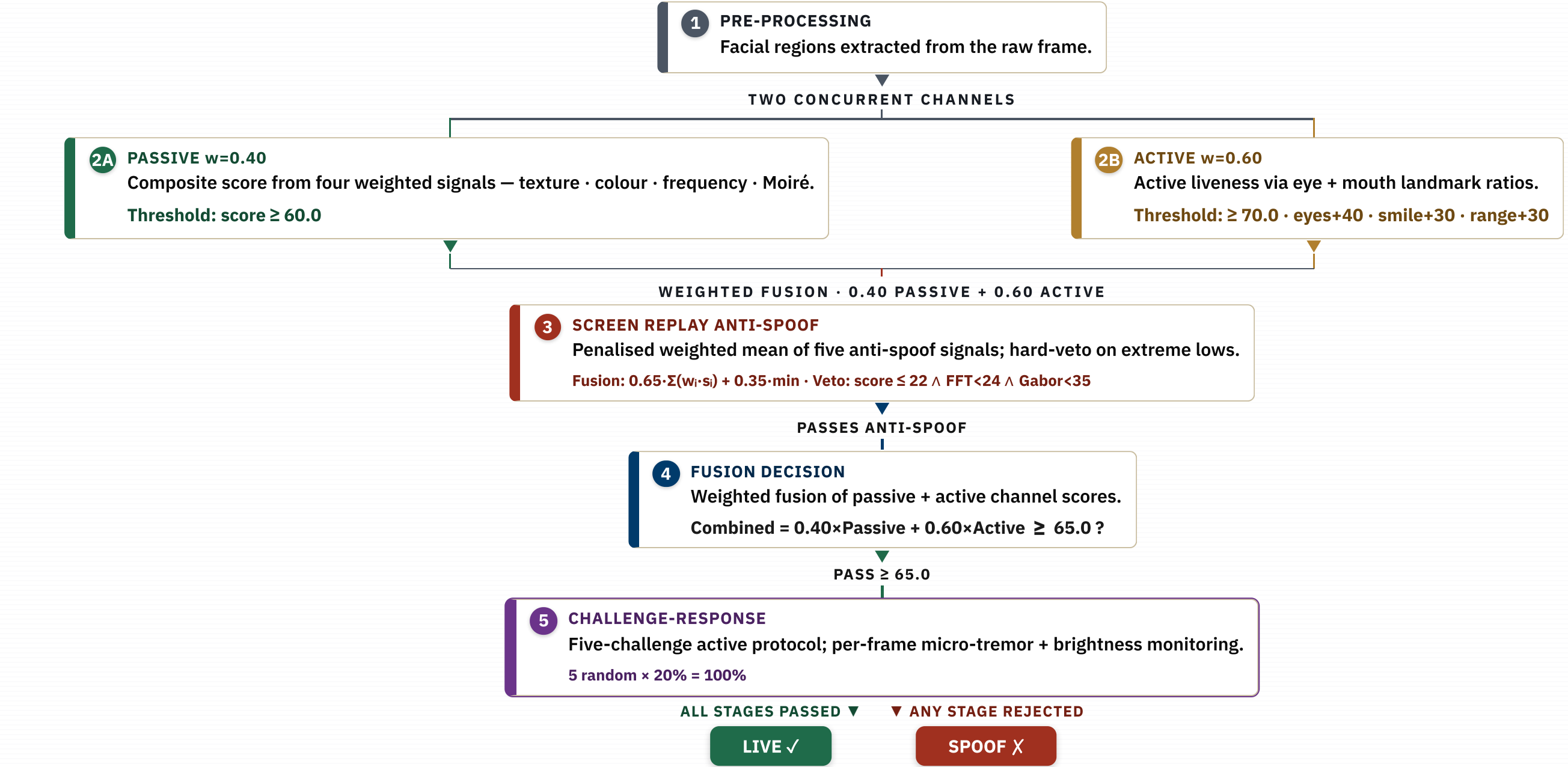
Production biometrics, zero GPU

The full face pipeline runs CPU-only on one commodity server (Hetzner CX43). Self-host with a single docker compose up .



5-Stage Hybrid Liveness Detection

Mitigates presentation attacks within FIVUCSAS - pre-process -> 2 concurrent channels -> anti-spoof -> fusion -> challenge.



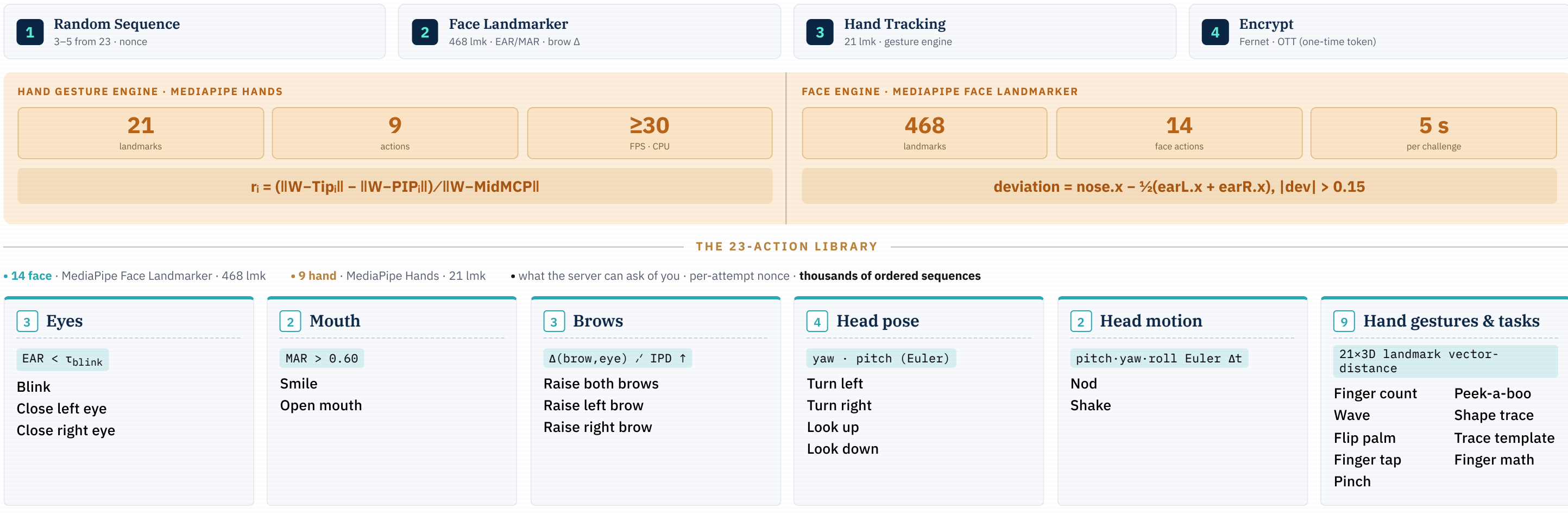
03 Benchmarking

FEATURE	FIVUCSAS	AUTH0 / OKTA	AWS REKOG.	AZURE FACE	APPLE FACE ID	FACETEC
Active Liveness	✓	X	✓	✓	✓	✓
Passive Liveness	✓	X	X	optional	✓	✓
On-Device Liveness	✓ active	X	X cloud	X cloud	✓ full	hybrid
OAuth 2.0 / OIDC	✓	✓	X	X	X	X
NFC Document	✓	X	X	X	X	✓
Composable MFA	✓ 10 factors	6 factors	X	X	X	X
Voice Biometrics	✓	X	X	X	X	X
Self-Hosted	✓	X	X	X	device only	✓ (SDK)
Multi-Tenant SaaS	✓	✓	X	X	X	X
No GPU Required	✓	n/a	cloud GPU	cloud GPU	neural engine	CPU/GPU

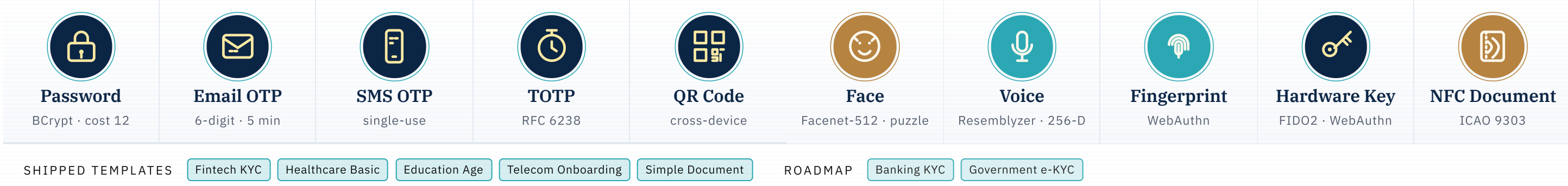
04 Key Innovations

The Biometric Puzzle

At each verification the server draws **3-5 random actions** from a 23-action library; the client performs them while landmarks are scored frame-by-frame under a strict **temporal contract**. Pre-recorded video, deepfake injection and replay all fail, because the action set is unpredictable and **timestamped per attempt**.



Ten Composable Authentication Factors



Reading the chip in your wallet

FIVUCSAS reads the contactless chip in **Turkish national ID cards** and **biometric passports** — documents following the **ICAO 9303** standard, now used by 150+ issuing states. The chip carries a signed photo, MRZ and a tamper-evident hash of every data group.

- **Mobile, native NFC**, Android client (Kotlin Multiplatform) runs a custom **ICAO 9303** reader on BouncyCastle — performs **BAC**, reads **DG1 / DG2** (MRZ + chip photo) and verifies the signed document hash (passive authentication).

05 Results

We ran a **thorough, hands-on evaluation** across recognition accuracy, cross-dataset robustness, real-time performance, liveness, and resistance to attacks. Tested on **1,342 enrolled face images across 100 identities** and **12,062 verification pairs** across three independent public benchmarks (LFW, AgeDB-30, CFP-FP).

PRODUCTION-GRADE · CPU-ONLY · MULTI-TENANT	99.43% FACE AUC	<0.5s P95 LATENCY	0 GPU REQUIRED	10/10 MFA FACTORS
A full identity platform measured end-to-end				

Face Recognition 0.9943 AUC LFW · 5,600 pairs · FaceNet-512 EER 2.93 % equal-error rate FAR 0.27 % @ threshold 0.45 TAR 95.6 % genuine accept rate STATE-OF-THE-ART	Cross-Dataset Robustness 0.9845 AUC CFP-FP · 1,378 pairs Pose Frontal -> 90° profile AgeDB-30 AUC 0.9475 (30-yr gap) Scale 100 identities · 12 k+ pairs GENERALISABLE	OAuth & Identity API ~66ms OIDC Discovery (warm) · production-measured JWKS ~62 ms · 6 live URLs all up Endpoints 208 REST · 29 controllers MFA 10 / 10 factors implemented PRODUCTION-LIVE
Production Pipeline ~410ms P95 end-to-end verification latency P50 ~380 ms · P99 ~450 ms Host Hetzner CX43 · 8 vCPU · 16 GB · 0 GPU Target <1.500 ms ✓ PASS SUB-SECOND	End-to-end face verification on commodity CPU ~410ms Median (P50) ~380ms P95 / PASS ~410ms P99 ~450ms Liveness PAD ~114ms Vector search ~345ms	

BY THE NUMBERS

An entire identity platform on commodity CPU

82 FLYWAY MIGRATIONS V0 -> V75 · production-applied schema evolution	208 REST ENDPOINTS 29 controllers · Identity Core API	41 DATABASE TABLES + 12 audit partitions · @Filter · pgvector IVFFlat
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06 Conclusion & Future Work

FIVUCSAS shows that **production-grade, KVKK / GDPR-compliant biometric authentication** can run on ordinary, CPU-only hardware, and ships both as **self-hostable open-source** and as a **managed SaaS**.

A. Harder problems solved

- **CPU-only ML** at sub-second latency on one 8-vCPU host.
- **Frozen OIDC contract** while ML iterates weekly.
- **Custom ICAO-9303 BAC reader** on BouncyCastle.
- **Tenant MFA flows** by JSON + Hibernate @Filter, no redeploy.
- **Spoof Detector:** standalone session-based passive PAD (13 analyzers · ISO/IEC 30107-3 harness · paper + public demo).

B. Future work

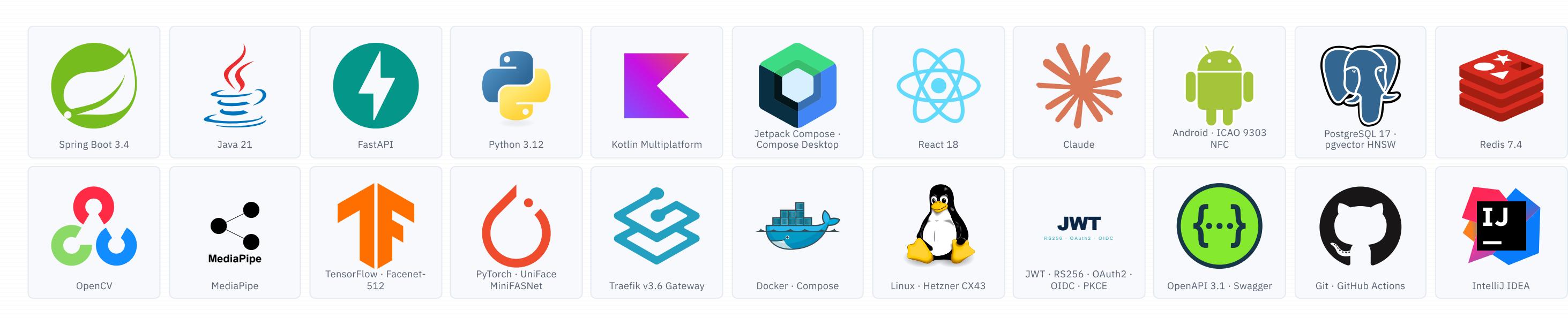
- **Voice upgrade.** Resemblyzer -> ECAPA-TDNN with replay-detection.
- **Edge AI build.** Raspberry Pi / Jetson endpoints on the same OIDC contract.
- **ISO/IEC 30107-3 conformance.** Accredited PAD audit.

Register once. Verify everywhere. Every time.

THE DOMAIN MAP: EVERY LINK BEHIND ONE QR

	fivucasas.com	Landing · live demo	verify.fivucasas.com	OAuth2 / OIDC hosted login
	app.fivucasas.com	Admin dashboard · independent UI	api.fivucasas.com	REST · OpenAPI 3.1
	demo.fivucasas.com	Sample scenario · functional test	status.fivucasas.com	API & server uptime / health
	links.fivucasas.com	QR aggregator (scan target)	github.com/Rollingcat-Software	MIT-licensed monorepo

07 Technologies Used



08 References

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